Day 9 — Routing Fundamentals & Routing Tables

Now that you understand **IP addressing, subnet masks, same/different networks**, we can move to the next topic in our schedule:

> **How does a router actually decide where to send an IP packet?**

This is the main question of Day 9.

---

1. First: What is Routing?

Routing means:

> **Choosing a path for an IP packet to reach its destination network.**

Example:

```
PC-A                Router                 PC-B
192.168.1.10        R1                     192.168.3.20
     |                 |                          |
     |                 |                          |
     └────────────────►|─────────────────────────►
```

PC-A wants to send data to:

```
192.168.3.20
```

PC-A knows:

```
192.168.3.20
```

is **not in its local network**.

So it sends the packet to its:

```
Default Gateway
```

The router receives it and asks:

> "Where should I send this packet next?"

That decision is **routing**.

---

2. Routing Happens at Layer 3

Remember our OSI model:

```
Layer 7  Application
Layer 6  Presentation
Layer 5  Session
Layer 4  Transport       TCP / UDP
Layer 3  Network        ← ROUTING
Layer 2  Data Link      Ethernet / MAC
Layer 1  Physical
```

The router's main forwarding decision is a **Layer 3 / IP decision**.

The router looks primarily at:

```
Destination IP
```

and uses its:

```
Routing Table
```

---

3. What is a Routing Table?

A routing table is basically a list of:

```
Destination Network
        +
Where to send it
```

For example:

```
Destination       Next Hop       Interface
------------------------------------------------
192.168.1.0/24    directly        eth0
192.168.2.0/24    10.0.0.2        eth1
192.168.3.0/24    10.0.0.3        eth1
0.0.0.0/0         10.0.0.1        eth1
```

Don't worry about every column yet.

The important idea is:

```
Destination IP
      ↓
Routing Table
      ↓
Select route
      ↓
Forward packet
```

---

4. Simple Example

Suppose router R1 has:

```
                 R1
             ┌──────────┐
             │          │
             │  Router  │
             │          │
             └──────────┘
               |      |
               |      |
        192.168.1.0   192.168.2.0
```

Its routing table might contain:

```
192.168.1.0/24 → eth0
192.168.2.0/24 → eth1
```

Now packet arrives:

```
Destination IP:
192.168.2.50
```

Router checks:

```
192.168.2.50
     ↓
Matches 192.168.2.0/24
     ↓
Use eth1
```

Then:

```
Router
  |
  | eth1
  ▼
192.168.2.50
```

---

5. The Router Does NOT Ask "Which Exact IP?"

This is important.

The routing table usually contains **networks/prefixes**, not one entry for every host.

Example:

```
192.168.2.0/24
```

means:

```
192.168.2.0
     ↓
192.168.2.1
     ↓
...
     ↓
192.168.2.254
```

So one route can cover an entire network.

---

6. Network Prefix

Consider:

```
192.168.2.0/24
```

The router interprets:

```
192.168.2
   |
   └── Network prefix
```

The host portion is:

```
last 8 bits
```

Therefore the route:

```
192.168.2.0/24
```

matches destinations such as:

```
192.168.2.10
192.168.2.50
192.168.2.100
192.168.2.200
```

---

7. Directly Connected Route

Suppose router has:

```
eth0 = 192.168.1.1/24
```

The router automatically knows:

```
192.168.1.0/24
```

is directly connected to `eth0` .

Conceptually:

```
Router
   |
   | eth0
   |
   ├───────────────
   |
192.168.1.0/24
```

Routing table:

```
192.168.1.0/24 → directly connected → eth0
```

So if destination is:

```
192.168.1.50
```

the router doesn't need another router.

---

8. What Does "Directly Connected" Mean?

It means:

> The destination network is attached directly to one of the router's interfaces.

Example:

```
             Router
        ┌──────┴──────┐
        |             |
      eth0          eth1
        |             |
        ▼             ▼
192.168.1.0/24   192.168.2.0/24
```

The router directly knows both networks.

---

9. Next Hop

Now consider:

```
PC-A          R1             R2           PC-B
 |             |              |             |
 |             |              |             |
 └────────────▶|─────────────▶|────────────▶|
```

PC-B belongs to:

```
192.168.3.0/24
```

R1 doesn't have that network directly connected.

Its routing table might say:

```
192.168.3.0/24 → 10.0.0.2
```

Here:

```
10.0.0.2
```

is the **next-hop router**.

So:

```
Destination:
192.168.3.20

Routing table:
192.168.3.0/24
        ↓
Next hop = 10.0.0.2
```

R1 sends the packet toward R2.

---

10. Full Routing Example

Let's build a complete network.

```
PC-A
192.168.1.10
     |
     |
     ▼
   R1
192.168.1.1
     |
```

```
      │ 10.0.0.1
      │
      ▼
    R2
 10.0.0.2
 192.168.3.1
      │
      │
      ▼
 PC-B
 192.168.3.20
```

PC-A wants:

```
192.168.3.20
```

---

11. Step 1 — PC-A Checks the Destination

PC-A:

```
192.168.1.10/24
```

Destination:

```
192.168.3.20
```

PC-A calculates:

```
My network:
192.168.1.0/24

Destination network:
192.168.3.0/24
```

Different.

Therefore:

```
Send to default gateway
```

---

12. Step 2 — PC-A Uses ARP

PC-A needs the MAC address of its gateway:

```
192.168.1.1
```

So it performs ARP:

```
PC-A
   |
   | Who has 192.168.1.1?
   ▼
Broadcast
   |
   ▼
Router R1
   |
   | I have it
   ▼
MAC-R1
```

Now PC-A knows:

```
Gateway IP:
192.168.1.1

Gateway MAC:
AA:AA:AA:AA:AA:AA
```

---

13. Step 3 — PC-A Creates the Packet

Layer 3:

```
┌────────────────────────────┐
│ Source IP                  │
│ 192.168.1.10               │
│                            │
│ Destination IP             │
│ 192.168.3.20               │
└────────────────────────────┘
```

Notice:

> Destination IP is PC-B.

Not the router.

---

14. Step 4 — Ethernet Frame

But the Layer 2 destination is:

```
Router's MAC
```

So:

```
┌────────────────────────────────┐
│ Ethernet                       │
│                                │
│ Source MAC = PC-A              │
│ Destination MAC = R1           │
│                                │
├────────────────────────────────┤
│ IP Packet                      │
│                                │
│ Source IP = 192.168.1.10       │
│ Destination IP = 192.168.3.20  │
└────────────────────────────────┘
```

This is one of the most important concepts from Days 7–9.

---

15. Step 5 — Router R1 Receives the Frame

R1 receives the Ethernet frame.

First:

```
Layer 2
   ↓
Ethernet destination MAC
   ↓
Is this frame for me?
   ↓
YES
```

R1 removes the Ethernet header/trailer and examines the IP packet.

---

16. Step 6 — R1 Looks at Destination IP

R1 sees:

```
Destination IP:
192.168.3.20
```

It checks its routing table.

Example:

```
Destination       Next Hop
--------------------------------
192.168.1.0/24    Direct
10.0.0.0/30       Direct
192.168.3.0/24    10.0.0.2
```

R1 finds:

```
192.168.3.0/24
```

Therefore:

```
Next hop:
10.0.0.2
```

---

17. Step 7 — R1 Creates a New Ethernet Frame

This is extremely important.

The IP packet is forwarded, but the **Layer 2 frame is rebuilt for the next link**.

Before R1:

```
PC-A → R1

MAC:
PC-A → R1
```

After R1:

```
R1 → R2

MAC:
R1 → R2
```

Visual:

```
           R1
           |
           |
Before     |      After
           |
PC-A ─────▶R1───────────▶ R2
 |          |              |
 |          |              |
MAC-A       MAC-R1         MAC-R2
```

More precisely:

```
Link 1:

Src MAC = PC-A
Dst MAC = R1
```

```
Link 2:

Src MAC = R1
Dst MAC = R2
```

18. But What Happens to IP?

The destination IP remains:

```
192.168.3.20
```

Conceptually:

```
First link:

IP:
192.168.1.10 → 192.168.3.20

MAC:
PC-A → R1
```

Second link:

```
IP:
192.168.1.10 → 192.168.3.20

MAC:
R1 → R2
```

Therefore:

```
IP = end-to-end destination
MAC = hop-to-hop delivery
```

⭐ **Memorize this.**

---

19. Step 8 — R2 Receives the Packet

R2 receives the frame.

It examines:

```
Destination IP:
192.168.3.20
```

R2 has:

```
192.168.3.0/24
```

directly connected.

Therefore:

```
192.168.3.20
      ↓
Directly connected network
      ↓
Send toward PC-B
```

---

20. Step 9 — R2 Needs PC-B's MAC

R2 needs the MAC address associated with:

```
192.168.3.20
```

It can use ARP on that local network.

Conceptually:

```
R2
 |
 | ARP:
 | Who has 192.168.3.20?
 ▼
Network
 |
 ▼
PC-B
 |
 | My MAC is BB:BB:...
 ▼
R2
```

---

21. Step 10 — Final Frame

R2 creates the final Ethernet frame:

```
┌──────────────────────────────┐
| Ethernet                     |
|                              |
| Source MAC = R2              |
| Destination MAC = PC-B       |
```

```
|                              |
├──────────────────────────────┤
| IP                           |
|                              |
| Source IP = PC-A             |
| Destination IP = PC-B        |
└──────────────────────────────┘
```

Then:

```
R2
 |
 | Ethernet
 ▼
PC-B
```

---

22. Complete Visualization

This is the diagram I want you to understand:

```
 PC-A                   R1                    R2                   PC-B
192.168.1.10          Router                Router              192.168.3.20
     |                   |                     |                     |
     |                   |                     |                     |
     |  Ethernet         |                     |                     |
     | MAC-A → MAC-R1    |                     |                     |
     |──────────────────▶|                     |                     |
     |                   |                     |                     |
     |                   | Routing Table       |                     |
     |                   |                     |                     |
     |                   | Destination         |                     |
     |                   | 192.168.3.20        |                     |
     |                   |       ↓             |                     |
     |                   | Next Hop R2         |                     |
     |                   |                     |                     |
     |                   |  Ethernet           |                     |
     |                   |  MAC-R1 → MAC-R2    |                     |
     |                   |────────────────────▶|                     |
     |                   |                     |                     |
     |                   |                     | Routing Table       |
     |                   |                     |                     |
     |                   |                     | 192.168.3.0/24      |
     |                   |                     |       ↓             |
     |                   |                     | Directly Connected  |
     |                   |                     |                     |
     |                   |                     | Ethernet             |
     |                   |                     | MAC-R2 → MAC-B       |
     |                   |                     |────────────────────▶|
     |                   |                     |                     |
```

---

23. The Three Different Decisions

This is the easiest way to organize everything you've learned.

PC-A first asks:

> **Is PC-B in my network?**

Uses:

```
IP + Subnet Mask
```

If different network:

> **Which router should I send to?**

Uses:

```
Default Gateway
```

Router asks:

> **Where should I forward this packet?**

Uses:

```
Destination IP
      ↓
Routing Table
      ↓
Longest Prefix Match
      ↓
Next Hop / Interface
```

---

24. Longest Prefix Match

Now let's connect this to the subnetting from Day 8.

Suppose router has:

```
10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
0.0.0.0/0
```

Destination:

```
10.10.20.50
```

All of these can potentially match:

```
10.0.0.0/8
       ↓
10.10.0.0/16
       ↓
10.10.20.0/24
```

Which one does the router choose?

```
/24
```

because it is the **most specific route**.

So:

```
Larger prefix length
        ↓
More specific
        ↓
Higher priority
```

---

25. What is the Default Route?

You will frequently see:

```
0.0.0.0/0
```

This means:

> "If nothing more specific matches, use this route."

Example:

```
Routing table

192.168.1.0/24 → eth0
```

Destination:

```
8.8.8.8
```

Does it match:

```
192.168.1.0/24 ? NO
192.168.2.0/24 ? NO
```

Then:

```
0.0.0.0/0
      ↓
Default route
```

---

26. Linux Routing Table

On your Ubuntu system, run:

```
ip route
```

You may see something like:

```
default via 192.168.1.1 dev wlp2s0
192.168.1.0/24 dev wlp2s0 proto kernel scope link src 192.168.1.20
```

Interpretation:

First line

```
default via 192.168.1.1
```

means:

```
Unknown destination
        ↓
Send to gateway
192.168.1.1
```

Second line

```
192.168.1.0/24 dev wlp2s0
```

means:

```
192.168.1.0/24
        ↓
Directly connected
        ↓
Use wlp2s0
```

---

27. Very Important Linux Command

You can ask Linux:

> "How would you route this IP?"

Use:

```
ip route get 8.8.8.8
```

For example, conceptually:

```
8.8.8.8
   ↓
Routing table
   ↓
Default route
   ↓
Gateway
   ↓
Interface
```

This is a very useful troubleshooting command.

---

28. Router vs Switch

You should now clearly distinguish them.

Switch	Router
Mainly Layer 2	Mainly Layer 3
Uses MAC addresses	Uses IP addresses
Uses MAC address table	Uses routing table
Forwards Ethernet frames	Forwards IP packets

Switch	Router
Connects devices within LAN/VLAN	Connects different IP networks
Learns MAC addresses	Learns/uses routes

Visual:

```
Switch:

MAC Address
     ↓
MAC Table
     ↓
Ethernet Frame
     ↓
Port



Router:

Destination IP
     ↓
Routing Table
     ↓
Next Hop / Interface
     ↓
New Layer-2 Frame
```

---

29. One Important Correction From Your Earlier Learning

You previously asked whether:

> "Router only does rework in Layer 2."

The better way to understand it is:

Router makes its forwarding decision at Layer 3, but it also has to process Layer 2 on each attached link.

For example:

```
Incoming:

Ethernet
   ↓
IP
   ↓
```

```
Routing decision
    ↓
New Ethernet
    ↓
Outgoing
```

So the router does **not** simply "work at Layer 2."

It:

```
Receives Layer-2 frame
        ↓
Processes Layer-3 IP packet
        ↓
Makes routing decision
        ↓
Creates outgoing Layer-2 frame
```

That's the exact model you should use in interviews.

---

30. ⭐ Day 9 Core Concept

Memorize this flow:

```
             IP Packet arrives
                    │
                    ▼
            Destination IP
                    │
                    ▼
             Routing Table
                    │
                    ▼
          Longest Prefix Match
                    │
          ┌─────────┴─────────┐
          │                   │
          ▼                   ▼
    Directly Connected      Next Hop
          │                   │
          └─────────┬─────────┘
                    ▼
              Output Interface
                    │
                    ▼
             ARP if required
                    │
                    ▼
          New Ethernet Frame
                    │
```

```
                    ▼
                 Next Hop
```

And remember this sentence for your interview:

> **A router examines the destination IP address, looks up the best matching route in its routing table using longest-prefix matching, selects an outgoing interface/next hop, and forwards the packet using a new Layer-2 frame on the next link.**

🧠 Day 9 Practice

Before moving to Day 10, try answering these without looking:

1. What is routing?
2. What is a routing table?
3. What is a directly connected route?
4. What is a next hop?
5. What is a default route?
6. What does `0.0.0.0/0` mean?
7. What is longest-prefix matching?
8. Why does a router change the Ethernet MAC addresses?
9. Does the destination IP normally change when a packet passes through a router?
10. What is the difference between a switch's MAC table and a router's routing table?
11. What does `ip route` show?
12. What does `ip route get 8.8.8.8` help you determine?

Once these are clear, Day 10 will be ICMP → `ping` → `traceroute`, where we'll use the routing concepts from Days 7–9 to understand exactly how `ping` and `traceroute` work packet-by-packet.